

Dual-Mode Cognitive Radio Network with Distributed Fusion and Self-Healing Communication Channel

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Abstract—This paper introduces a distributed, state-aware cognitive radio network architecture with a self-healing communication channel (CC) mechanism. Leveraging two operational modes: (1) *Interweave Mode* for standard communication over unoccupied channels, and (2) *Underlay Mode* to sustain CC functionality during primary user (PU) activity. Multiple fusion centers (cluster heads) among secondary users (SUs) are responsible for cooperative sensing and decentralized CC selection. Each cluster head stochastically accesses the underlay CC using a CSMA-CA-inspired randomized backoff to avoid collisions with other cluster heads. The proposed architecture enhances CC resilience and collision avoidance in dense cognitive radio environments.

Index Terms—Cognitive radio, dynamic spectrum access, common control channel, distributed coordination, self-healing networks

I. INTRODUCTION

The rapid growth of wireless technologies has intensified the demand for radio spectrum, yet vast portions of licensed bands remain underutilized. Cognitive radio networks (CRNs) emerged as a transformative solution, enabling dynamic spectrum access where unlicensed secondary users (SUs) opportunistically utilize idle frequency bands owned by licensed primary users (PUs). This paradigm promises to alleviate spectrum scarcity while maintaining strict adherence to PU priority.

Conventional CRN architectures often depend on centralized controllers or shared control channels to coordinate spectrum access. Centralized systems, while efficient, introduce critical vulnerabilities: reliance on infrastructure, single points of failure, and scalability constraints [2]. Semi-distributed designs decentralize decision making but remain tied to dedicated control channels, which are prone to congestion, jamming, and licensing complexities [2]. These limitations hinder deployment in dynamic, large-scale, or resource-constrained environments.

Fully distributed CRNs eliminate centralized control and dedicated coordination channels, empowering SUs to operate autonomously. Although this architecture improves scalability and resilience, uncoordinated spectrum access risks inefficiency [3]. Competing SUs may collide frequently, misjudge band availability, or endure prolonged outages during PU reappearances—challenges that degrade performance in real-

time applications such as industrial automation or emergency response networks.

This work proposes a self-organizing, cluster-based CRN framework designed to harmonize autonomy with intelligent coordination. SUs dynamically form clusters, each managed by an elected cluster head (CH) that facilitates localized decision-making. Unlike prior distributed models, the system integrates two key innovations:

Adaptive Cluster Coordination: CHs employ lightweight signaling to optimize band allocation within clusters, reducing contention.

Context-Aware Backoff Mechanisms: A tunable backoff algorithm adapts to real-time network conditions, balancing collision avoidance and rapid spectrum access.

II. SYSTEM ARCHITECTURE

A. Assumptions and PU Model

- A single primary user (PU) occupies the spectrum but permits low-power underlay communication.
- 20 frequency bands (including 1 underlay band)
- There is 25 total SUs, 5 SUs act as CHs.
- each CH expects 4 packets each time slot (1 packet from each SU)
- Communication channel operation switches modes to adapt to PU interference.

B. State Definitions

- **State 1 (Interweave):**

$$S_1 = \begin{cases} \text{CC: Free access (Interweave)} \\ \text{Data channels: Interweave} \end{cases} \quad (1)$$

- **State 2 (Underlay):**

$$S_2 = \begin{cases} \text{Emergency CC: Low-power TX (Underlay)} \\ \text{Data channels: Interweave} \\ \text{CC Selection: Cooperative (via CH)} \\ \text{Access Timing: Randomized CH slot backoff} \end{cases} \quad (2)$$

C. Cluster Coordination

- CHs aggregate reports, and select candidate CCs.
- A CH randomly chooses one CC from the list to propose as a new CC.

III. SYSTEM DESIGN

A. Randomized Backoff Algorithm with Collision Avoidance

The system implements a randomized exponential backoff algorithm for collision avoidance when multiple Cluster Heads (CHs) contend for underlay band access. Inspired by CSMA/CA principles, displaced CHs initiate the process with a small random backoff (2-6 slots) to prevent synchronized retries. Upon collision detection - simulated through a probabilistic model combining a 12% base chance with PU activity-dependent scaling - CHs adaptively double their backoff window (capped at 512 slots) while incorporating 0-24 slot jitter to maintain responsiveness under varying congestion levels. The mechanism features two critical recovery behaviors: successful underlay access resets the backoff counter entirely, while CHs exceeding 16 consecutive collisions enter temporary suspension, emulating practical hardware constraints. This balanced approach minimizes repeated collisions during high-PU-activity scenarios while ensuring fair access opportunities across all clusters.

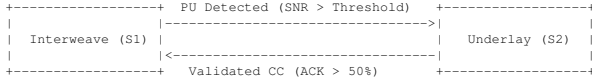
B. Control Packet Specifications

Type	Fields	Purpose
0x01	Type, Data	SU to CH Report
0x02	Type, Data	CH to SUs Report
0x03	Type, CC Candidates, Power Limit	Underlay Command
0x04	Type, Candidate ID, Quality Metric	CC Proposal
0x05	Type, Final CC, Sync Time	Transition Sync

TABLE I: Control packet formats

IV. STATE TRANSITION PROTOCOL

A. State Transition Diagram



B. Transition Algorithm

Algorithm 1 Distributed State Transition Logic (Per CH)

Input : Current state S_t , SNR_{CC} , ACK_{count} , candidate set $\mathcal{C}$, K

Output: New state S_{t+1}

if $S_t = S_1$ **and** $SNR_{CC} > Threshold\ dB$ **then**

Wait for random K time slot

$S_{t+1} \leftarrow S_2$

Broadcast CC Candidate(0x04) with selected $c \in \mathcal{C}$

end

else if $S_t = S_2$ **and** $ACK_{count} \geq 0.5N_{SU}$ **then**

Broadcast InterweaveSync(0x05)

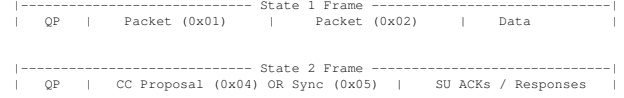
$S_{t+1} \leftarrow S_1$

Resume cooperative sensing

end

V. TIMING AND SYNCHRONIZATION

A. Frame Structure



VI. PERFORMANCE EVALUATION

A. Definitions

- **Packet Delivery Ratio (PDR):** The fraction of time slots where the Cluster Head successfully received all 4 packets from its Secondary Users. Only incremented in NORMAL state.

What it measures: Reliability of complete packet delivery.

$$PDR_i = \frac{\text{fullPacketSlots}_i}{T}$$

Where:

fullPacketSlots_i = Number of timeslots where CH i received all 4 packets

T = Total simulation slots (30,000)

- **Packet Loss Percentage:** Percentage of total expected packets that were not received due to CH being in outage states.

What it measures: Overall packet loss rate due to displacement/underlay operation.

$$\text{PacketLossPercentage}_i = \left(\frac{\text{totalPacketsLost}_i}{T \times K} \right) \times 100\%$$

Where:

$\text{totalPacketsLost}_i$ = Total packets lost by CH i

K = Number of SUs per CH ($K = 4$)

$T \times K$ = Total possible packets ($30,000 \times 4 = 120,000$)

- **Average Outage Duration:** Average length (in time slots) of each outage event when CH is displaced from its normal band.

What it measures: How long CH typically stays displaced during PU activations.

$$\text{AvgOutageDuration}_i = \begin{cases} \frac{\text{outageSlots}_i}{\text{outageEvents}_i} & , \text{outageEvents}_i > 0 \\ 0 & , \text{otherwise} \end{cases}$$

Where:

outageSlots_i = Total slots in DISPLACED or UNDERLAY states

outageEvents_i = Number of times CH i transitioned from NORMAL state

B. Simulation Results

As shown in Fig.1 below, the PDR is always higher than 80% regardless of the probability of activation of PU, showing the robustness of the system.

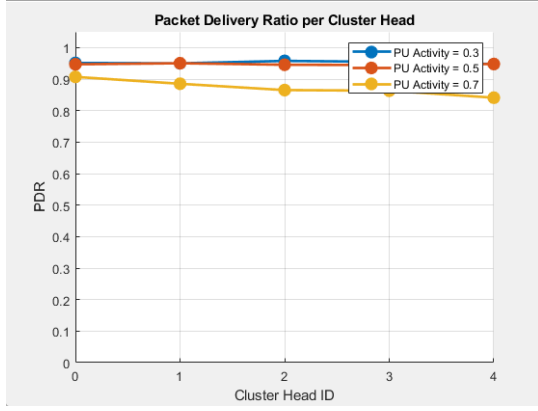


Fig. 1: Packet Delivery Ratio per Cluster Head.

As shown in Fig.2 below, the packet loss percentage is relatively low. Showing how the system keeps stable and reliable communication even under violent PU activity.

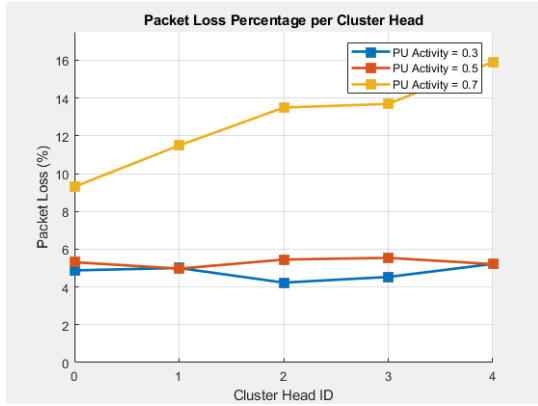


Fig. 2: Packet Loss Percentage per Cluster Head.

Fig.3 shows the outage duration when switching between states, outage durations increase with PU activity but remain relatively low due to adaptive backoff.

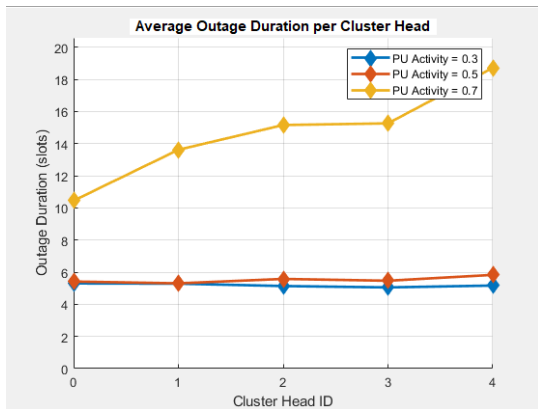


Fig. 3: Average Outage Duration per Cluster Head

Fig.4 illustrates how the CH behaves in a crowded network (PU activation probability of 0.7) and how it manages to stay

in the interweave state for as long as possible and minimizes the time in underlay state and in the displaced state (waiting to enter the underlay band).

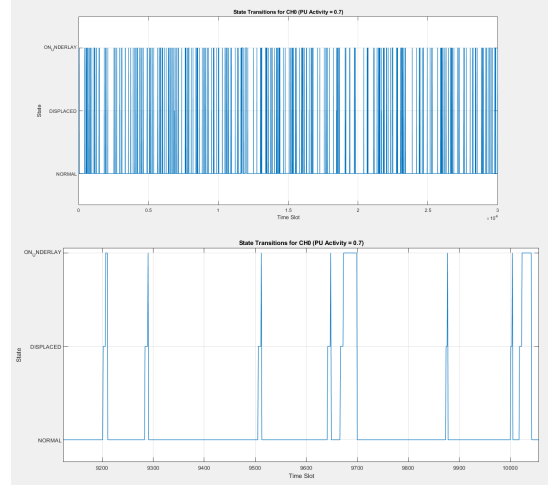


Fig. 4: State Transitions for CH0 (PU Activity = 0.7)

VII. CONCLUSION

The enhanced dual-mode cognitive radio architecture integrates distributed control via CHs and a CSMA-CA-inspired backoff for underlay CC contention. Key benefits include:

- **Self Recovery:** CHs independently detect CC failure and initiate recovery.
- **Collision Avoidance:** Randomized slot-based access reduces CH interference in underlay mode.
- **Reliable CC Maintenance:** Decentralized CC election ensures robust operation in dynamic spectrum environments.

VIII. FUTURE WORKS

A. Bootstrap and Initialization

When a secondary user (SU) first activates, it executes the following procedure to discover or establish a cluster:

Algorithm 2 SU Bootstrap Protocol

Input : Available spectrum bands $\mathcal{B}$, Beacon timeout T_b , Energy detection threshold γ

Output: Cluster membership status (CH or member)

Scan all channels $c \in \mathcal{B}$ for beacon signals

Measure received power P_c on each channel for duration T_b

if $\exists c$ where $P_c > \gamma$ **and** valid beacon detected **then**

 Join cluster using parameters from beacon

 Sync to CH's frame structure

 Participate in cooperative sensing

end

else

 Perform energy detection on all candidate channels

 Select channel $c^* = \min_c P_c$ as new CCC

 Broadcast periodic beacon packets on c^*

 Assume CH role

end

Beacon Packet Structure: The beacon contains essential cluster synchronization information:

$$\text{Beacon} = \langle \text{CH ID}, \text{CC}, t_{\text{sync}}, N_{\text{members}}, \text{Sensing params} \rangle \quad (3)$$

Field	Size	Description
Header	2 bytes	Always 0x06 (Beacon identifier)
CH ID	4 bytes	Cluster head unique identifier
CC	2 bytes	Current control channel index
t_{sync}	8 bytes	Next frame start timestamp
Cluster Size	2 bytes	Current number of member SUs
Sensing Threshold	2 bytes	Energy detection threshold (dBm)
CRC	2 bytes	Packet integrity check

TABLE II: Beacon packet structure (Type 0x06)

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